Reverse-GNSS Spacecraft Multi-Observable EKF Report

Scenario: **GEO-1**

Reverse-GNSS EKF simulator, validation version 2.02

Generated 2026-08-19 10:05:37

Controlled synthetic reverse-GNSS scenario.

1 Goal and Scenario

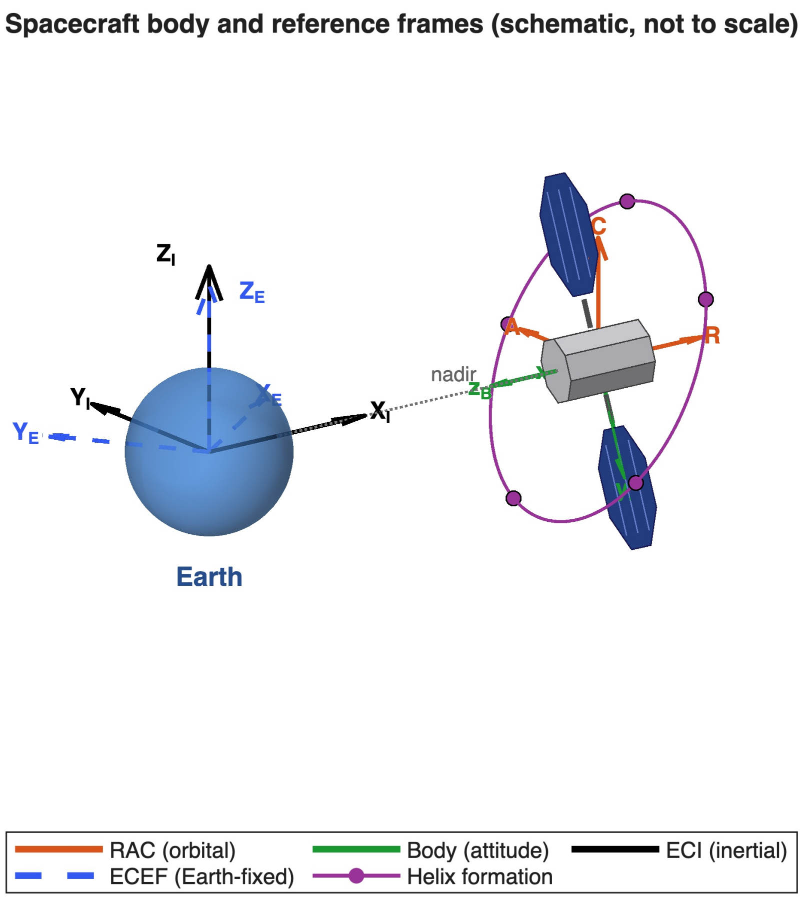
Ground transmitters send signals UP to the spacecraft, which is the reverse of normal GNSS. This run asks a single question: how well can one GEO-class spacecraft work out its own orbit, receiver clock and attitude from those signals alone? The estimator is compared against the synthetic truth used to generate the measurements, and the sections below give the geometry, the noise assumptions, and how consistent the residuals are.
Run length 1.00 h (3600 s) at 1.0 s sampling. Every number here is valid for this synthetic scenario only; it is not a PPP-grade or real-data performance claim.

Metric	Value
Baseline requirement	Single GEO asset reverse-GNSS estimation scenario
Orbit class	GEO (geostationary)
Space assets	6
Ground transmitters	5
On-board receivers	4
Enabled measurement families	single-frequency code, float carrier ambiguity EKF, Doppler, ground reciprocal time transfer
Simulation duration	3600 s (1.00 h)
Validation version	2.02

The ground-to-space time-transfer mode is `firstOrderReciprocal`. It supplies calibrated receiver–tower clock-difference rows; it is not a primitive four-timestamp event simulation.

1.1 Coordinate Frames and Units

Frame	Use
ECI	Orbit propagation and the radial/along-track/cross-track post-processing reference.
ECEF	Ground transmitter coordinates, receiver phase centres, line-of-sight geometry, and measurement rows.
RAC	Radial, along-track, and cross-track decomposition of the estimate-minus-truth position error.
Body	Attitude error-state and antenna lever-arm interpretation.
Clock units	Clock bias in metres and seconds; clock drift in m/s and s/s.



Space Asset and Reference Frames. Stylised schematic (not to scale) of the primary space asset, an octagonal bus with two octagonal solar panels, with the ECI (inertial), ECEF (Earth-fixed), RAC (radial / along-track / cross-track orbital) and body-attitude frames, and, for a swarm, the helix formation. For an equatorial GEO the cross-track axis coincides with the ECI spin axis. Generated by make_utils_figures.m into output/utils/.

1.2 Clocks

A one-way pseudorange sees the DIFFERENCE of two clocks, so both are listed here with the sign each carries into the measurement.

Item	This run
Receiver clock (spacecraft)	`CESIUM1`, stochastic (Brown-Hwang two-state, power-law h coefficients)
Transmitter clock (ground)	`RUBIDIUM`, stochastic (Brown-Hwang two-state, power-law h coefficients)
Signs in the pseudorange	b_{rx} enters **positive** (adds range), b_{twr} enters **negative** (subtracts). Troposphere T is positive for code and carrier alike. Ionosphere I_f is positive for code (group delay) and negative for carrier (phase advance). The carrier ambiguity B_ϕ is a float value in metres.
Clock states in the EKF	RX only: receiver clock bias and drift. Tower clocks are not states; they enter through external corrections
Clock architecture	spacecraft receiver clock only
Clock gauge	`externalTowerCorrections`. The datum comes from outside the filter, as broadcast tower-clock corrections

Why a gauge is needed at all: a one-way pseudorange only ever measures $b_{rx} - b_{twr}$, so the two biases cannot be separated without an external datum. The gauge row above is what supplies it.

1.3 State Vector

The filter is an error-state EKF. The 14 base states are grouped below; optional blocks are appended when active. Index ranges are computed from the active filter configuration, not hard-coded.

State group	Indices	Dim	Unit	Description
position	x[1:3]	3	m	spacecraft position error (RAC/ECEF as configured)
velocity	x[4:6]	3	m/s	spacecraft velocity error
attitude	x[7:9]	3	rad	small-angle body attitude error
angular rate	x[10:12]	3	rad/s	body angular-rate error
receiver clock bias	x[13]	1	m	receiver clock bias (positive sign)
receiver clock drift	x[14]	1	m/s	receiver clock drift
float ambiguities	x[15:34]	20	m	one float carrier ambiguity per active tower/receiver/signal arc
zenith wet delay	x[35:39]	5	m	per-tower zenith wet delay residual
total	x[1:39]	39	n/a	active EKF state dimension

1.4 Carrier Signals

Signal	Carrier frequency	Wavelength	Enabled
L1	24.1250 GHz	1.24 cm	yes
L2	5.8000 GHz	5.17 cm	yes

The labels `L1` and `L2` are catalogue names, not a claim that the run used the GPS L-band: a scenario may retune either band, and the frequencies above are the ones actually simulated. The wavelength is c/f and sets the carrier-phase cycle-to-metre conversion; the ionosphere scales as $1/f^2$ between them.

1.5 Measurement Model Equations

$$
\begin{aligned}
\rho_{\text{code}} &= \rho_{\text{geom}} + b_{\text{rx}} - b_{\text{tx}} + T + I_{\text{code}} + \Delta_{\text{sagnac}} + \Delta_{\text{rel}} + \Delta_{\text{ant}} + B_{\text{code}} + M + \epsilon_\rho \\
\Phi &= \rho_{\text{geom}} + b_{\text{rx}} - b_{\text{tx}} + T + I_{\text{carrier}} + \Delta_{\text{sagnac}} + \Delta_{\text{rel}} + \Delta_{\text{ant}} + N_{a,r,s} + B_\phi + M + \epsilon_\phi \\
D &= \dot{\rho} + \dot{b}_{\text{rx}} - \dot{b}_{\text{tx}} + \dot{\Delta}_{\text{corr}} + \epsilon_D \\
\nu &= z - h(\hat{x}^-)
\end{aligned}
$$

$$
\begin{aligned}
\rho_{\text{ISL}} &= \|\mathbf{r}_{\text{sc}} - \mathbf{r}_j\| + b_{\text{rx}} - b_j + \epsilon_{\text{ISL}} \\
D_{\text{ISL}} &= \mathbf{u}_{\text{sc},j}^\top(\mathbf{v}_{\text{sc}} - \mathbf{v}_j) + \dot{b}_{\text{rx}} - \dot{b}_j + \epsilon_{D,\text{ISL}}
\end{aligned}
$$

The inter-satellite link (ISL) rows apply only when the multi-asset swarm scenario is enabled: j is a neighbouring space asset and $\mathbf{u}_{\text{sc},j}$ is the inter-asset line of sight.

$$
P_{\text{IF}} = \alpha\, P_{L1} + \beta\, P_{L2}
$$

Term	Expression	Meaning
geometric range	$\rho = \|\mathbf{r}_{sc} + \mathbf{C}_{BI}\mathbf{l}_{a,B} - \mathbf{r}_{twr}\|$	Phase-centre to tower range
receiver clock	$+b_{rx}$ [m] (POSITIVE sign)	Shared spacecraft RX clock bias
tower clock	$-b_{twr}$ [m] (NEGATIVE sign)	Ground transmitter clock bias
troposphere	$+T$ (code and carrier, POSITIVE)	Slant wet+dry delay, same sign
iono code	$+I_f$ (POSITIVE for code)	First-order group delay

iono carrier	$-I_f$ (NEGATIVE for carrier)	First-order phase advance
float ambiguity	$+B_\phi$ [m] (L1 only)	L1 carrier cycle ambiguity, float
measurement noise	$\nu \sim N(0, R)$	Code / carrier / Doppler noise
IF coefficient α	$\alpha = 1.0613$	L1 weight in the ionosphere-free code combination
IF coefficient β	$\beta = -0.0613$	L2 weight; from the RESOLVED band pair, not the L-band constants

1.6 Measurement Noise and Error Budget

Measurement model	**Value**
Ground observables	
Code pseudorange	enabled
Carrier phase	enabled
Doppler	enabled
Code noise model	cn0
Code σ at 45 dB-Hz	0.3 m
C/N$_0$ base / elevation gain	45 dB-Hz / 6 dB
Carrier phase σ	0.01 m
Doppler σ	0.0424 m/s
Measurement covariance floor	0.01 m
Shared-product covariance	enabled
Shared transmitter-clock covariance	enabled
Ground two-way time transfer	
Two-way time transfer	enabled
Two-way time transfer σ	0.03 m
Inter-satellite link (one-way, EKF rows)	
ISL code σ	0.015 m
ISL carrier σ	0.002 m
ISL Doppler σ	0.02 m/s
Beamforming	
Coherence criterion	$\lambda/20$ (18.0° RMS phase, -0.43 dB loss)
Ground beam-pointing lock	enabled
Pointing-lock towers	3
Pointing-lock spot σ	500 m
Pointing-lock minimum elevation	10 deg

Error budget (truth-side injection)	**Value**
Code thermal σ	0.3 m
Carrier phase σ	0.01 m
Doppler σ	0.0424 m/s
Tower clock product σ (bias)	0.1 m
Tower clock product σ (drift)	0.001 m/s
Multipath model	coloured Gauss-Markov
Multipath σ (steady state, L1 zenith)	0.3 m
Multipath correlation time	60 s
Multipath elevation exponent	1
Troposphere model	localWeatherGM
Troposphere zenith delay	from per-tower weather (Saastamoinen/Davis)
Troposphere wet σ (steady state)	0.04 m
Troposphere correlation time	1.08e+04 s
Troposphere model residual σ	0.02 m
Ionosphere model	tecGaussMarkov
Ionosphere diurnal VTEC (day / night)	30 TECU / 6 TECU
Ionosphere TEC residual σ (steady state)	0.3 m
Ionosphere correlation time	600 s
Ionosphere model-side correction	klobuchar
Higher-order ionosphere	enabled
Scintillation model	conker
Scintillation code σ (L1)	0.3 m
Hardware delay σ	0.05 m
Code inter-frequency bias (truth L1 / L2)	0.3 m / 0.45 m
Receiver / transmitter clock process-noise drives	defined in the clock model (Appendix)

1.7 Starting Positions

Type	**Name**	**Frame**	**Coord 1**	**Coord 2**	**Coord 3**
Spacecraft	GEO-1	Geodetic (WGS84)	Lat 0.0000 deg	Lon 23.0000 deg	Alt 35786.0 km
Ground tower	Tenerife	Fixed geodetic	Lat 28.30 deg	Lon -16.50 deg	Alt 0.0 m
Ground tower	Stockholm	Fixed geodetic	Lat 59.30 deg	Lon 18.10 deg	Alt 0.0 m
Ground tower	Hartebeesthoek	Fixed geodetic	Lat -25.90 deg	Lon 27.70 deg	Alt 0.0 m
Ground tower	Bengaluru	Fixed geodetic	Lat 13.00 deg	Lon 77.60 deg	Alt 0.0 m
Ground tower	Libreville	Fixed geodetic	Lat 0.04 deg	Lon 9.45 deg	Alt 0.0 m

Core geometry and signals

Component	**Status**	**Note**
Ground segment geometry	Enabled	5 transmitting towers, 4 receive antennas.
L1+L2 dual-frequency signals	Enabled	L1+L2; IF combination available.
Carrier phase in EKF	Enabled	Gate is measurements.carrierMode = 'ekfFloat' (carrierPhase.enable is a fallback only).
Doppler in EKF	Enabled	model: frame-consistent range-rate

Clock and covariance

Component	Status	Note
Receiver clock (spacecraft)	Enabled	CESIUM1 oscillator; h-coefficients from Winkel (2003) Table 2.1.
Tower clock product correction	Enabled	External noisy product correction (synthetic broadcast tower-clock product (noisy)).
Tower clock product covariance	Enabled	R-inflation from product age and drift uncertainty.
Joint tower clock EKF	Disabled	mode: spacecraft receiver clock only; gauge: external tower-clock corrections.
ZWD / troposphere EKF state	Enabled	mode: Per tower zwd
Empirical RTN accelerations	Disabled	Disabled; force-model error is left unmodelled (biases the state, not P).
Per-tower TX code bias EKF	Disabled	0 estimated tower code-bias states.

Atmosphere and propagation

Error source	Truth in z	Model in h	Noise in R	What actually reaches the filter
Troposphere	●	●	●	Injected; residual absorbed by the per-tower ZWD EKF state.
Ionosphere	●	●	●	First order: SURVIVES on code rows, cancelled on carrier rows (L1/L2 IF combination). Remainder absorbed by the per-tower slant EKF state.
Light-time / Sagnac	●	●	○	Iterative one-way light-time (iterativeOneWay); Sagnac subsumed by the geometric Earth rotation, so a separate Sagnac term would double-count.
Shapiro delay (relativistic range)	●	●	○	Applied identically to truth and model: the residual is ZERO. Nothing about this effect reaches the filter.
Relativistic clock offset	●	●	○	Applied to truth and model: residual is ZERO.

Multi-asset scope (6 spacecraft): the effects above are applied to the chief's tower links. Each secondary satellite receives a divergent uplink atmosphere only when `multiAsset.towerSecondary.atmosphere` *is enabled (this run: off), and truth-side per-satellite dynamics only when* `injectTruthSideDynamics` *is enabled (this run: off). Inter-satellite links carry no atmosphere.*

Carrier, ambiguity and attitude

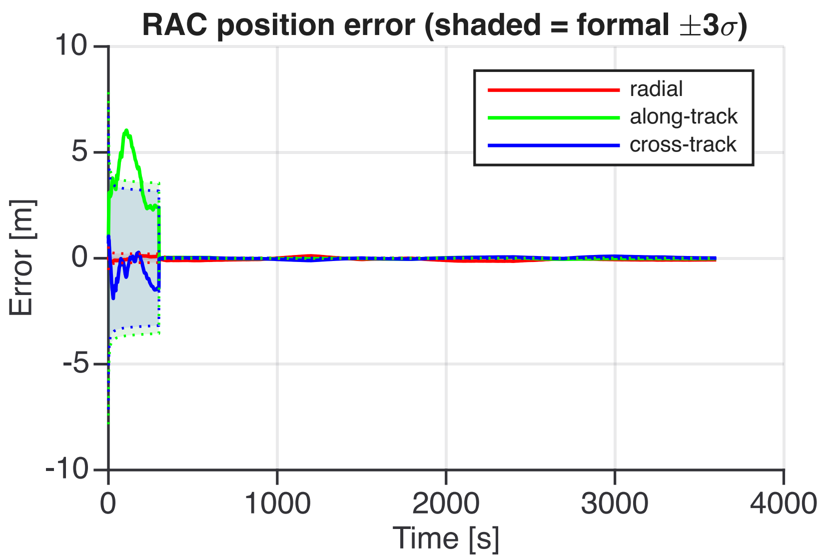
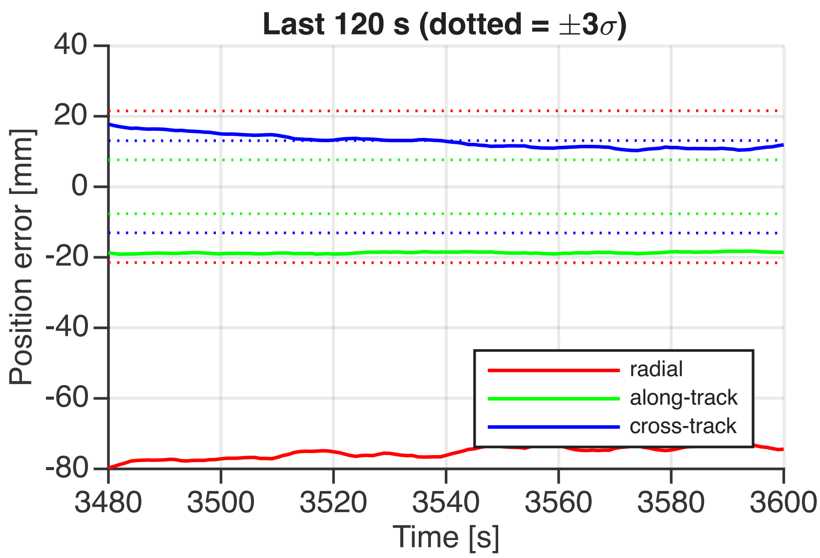
Component	Status	Note
Carrier L1 float rows in EKF	Enabled	Float ambiguity EKF, raw L1.
Carrier L2 float rows in EKF	Enabled	Float ambiguity EKF, raw L2 (L1+L2 active).
Carrier slip guards + arc sep	Disabled	n/a
Code IF rows in EKF	Disabled	Gate is measurements.codeMode = 'singleFrequency'.
Carrier IF rows in EKF	Enabled	Gate is CarrierIonoFreeRowBuilder.combineStatus (enable AND useInEkf AND nCarrierSignals¿1); the signal-count clause is now in the code, not just this note.
Raw carrier integer fixing	Disabled	Disabled; guarded raw-carrier fixing available, not active in this run.
Baseline attitude AR	Disabled	Not active; requires carrier float + 4rx + attitude EKF + diffAtt mode.
Attitude EKF (spacecraft)	Enabled	parameterisation: quaternion error-state
Diff. carrier att. calibration	Disabled	n/a

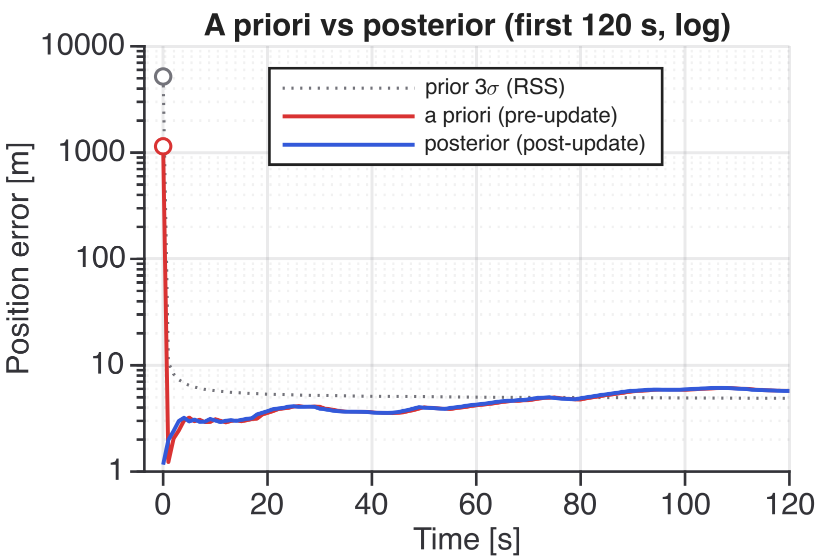
Antenna, hardware and unsupported

Component	Status	Note
Antenna PCO (truth)	Enabled	Active in this run.
Antenna PCV (truth)	Enabled	Gate is effects.antenna.pcvModel = 'toy'; 'none' yields an identically zero correction whatever the truth flag says.
Hardware delay (truth)	Enabled	Active in this run.
Multipath (truth)	Enabled	Time-correlated (coloured Gauss-Markov) code multipath.
LAMBDA / MLAMBDA	Disabled	Available on this branch but switched off.
Carrier IF integer fixing	Not impl.	Explicitly unsupported; the IF ambiguity is not an integer.
ANTEX / SP3 / CLK parsers	Not impl.	Synthetic constants only; no file-based corrections.
PPP-grade processing	Not impl.	Not implemented.

2 State Estimation Validation

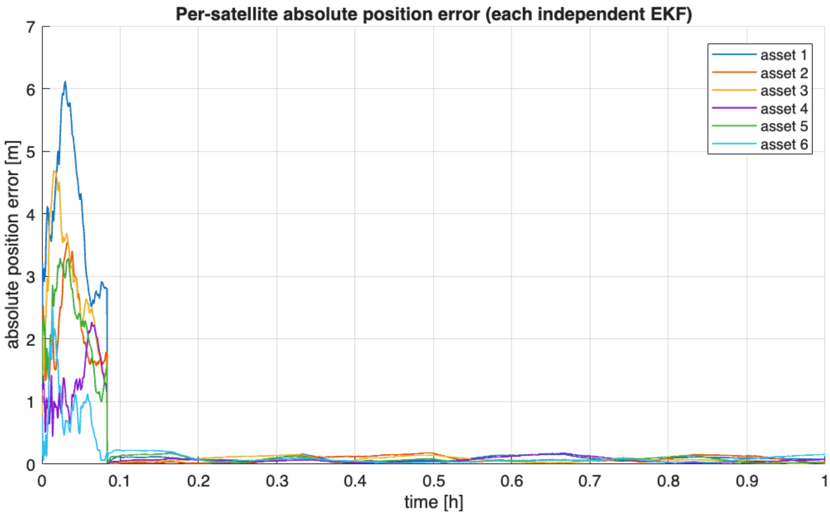
The plots compare EKF state estimates with truth using deterministic truth differencing. No measurement noise is injected unless the noise toggle is enabled. Position error is shown in the RAC (radial / along-track / cross-track) orbital frame as estimate minus truth. The RAC basis is built from the truth position and (inertial) velocity.

Plot	**Description and statistical approach**
	**Position Error, RAC Frame (estimate minus truth)** Position error decomposed into radial, along-track, and cross-track components in the orbital frame defined by the truth position and inertial velocity. Sign convention is estimate minus truth; the y-axis unit is chosen automatically. The shaded bands are the filter's formal $\pm 3\sigma$. They state PRECISION, not accuracy: they are built from H, R and Q alone and carry no term for unmodelled systematics, so the error can leave the band without the filter being wrong about what it computes.
	**Position Error (RAC), Final Zoom** RAC position error over the final 120 s. Confirms convergence. RMS over this window: radial 0.0754 m, along-track 0.0187 m, cross-track 0.013 m.



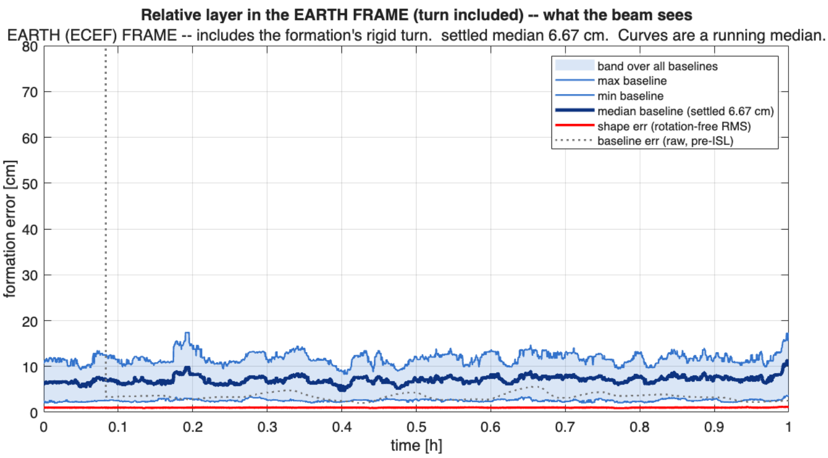
Initial Transient, A Priori vs Posterior

Every other position plot here is POSTERIOR, written after the epoch's update, so the configured initial offset never appears in them. The a priori curve is the state before the update, so its first point IS that initial error, and the gap between the two curves at the first epoch is the correction the opening update applies. Dotted line: a priori formal 3-sigma (RSS over the three axes). Log axis.



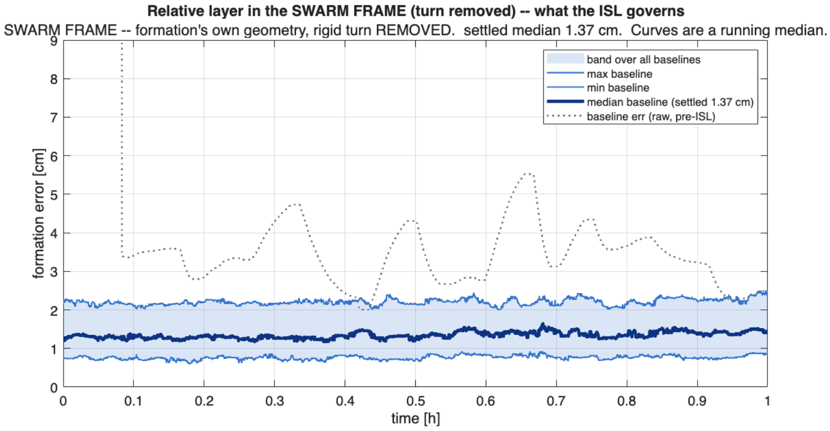
Federated Swarm, Per-Satellite Absolute Position Error

Each satellite estimates its own absolute state with an independent single-asset EKF (no chief, no shared covariance). Under the GEO radial–clock degeneracy every absolute solution is wall-limited (common mode), so the per-satellite errors settle at the metre level rather than to zero.



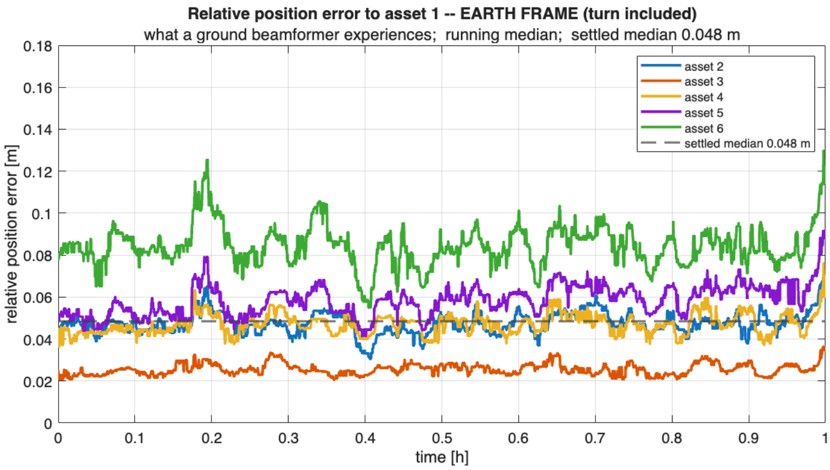
Federated Swarm, Relative Layer (ISL Shape / Clocks)

This legacy read-only relative-layer diagnostic creates synthetic range observations from stored truth trajectories plus configured noise. It is rigid-motion-blind and does not update a per-asset EKF or covariance.



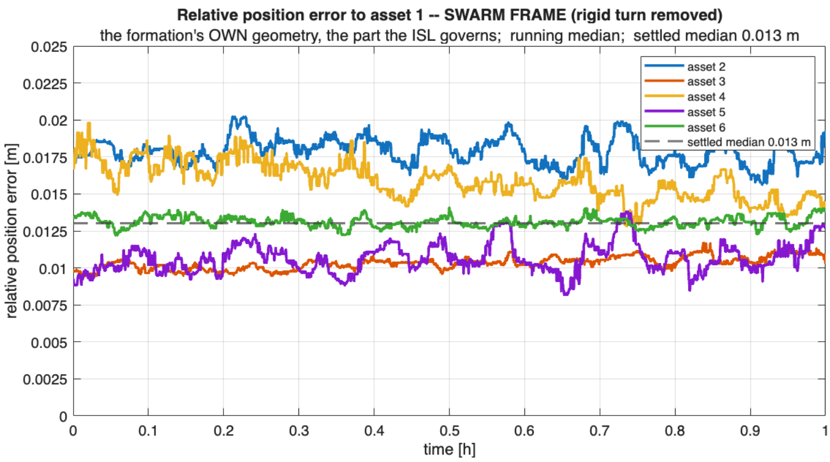
Federated Swarm, Relative Layer in the SWARM FRAME

The SAME relative layer as the plot above, but with the formation's rigid turn removed, so what remains is the formation's OWN internal geometry error. This is the part inter-satellite ranging governs. Comparing the two plots separates the two failure modes: this one is set by crosslink delay calibration, the Earth-frame one adds the formation's orientation, which no range observable can constrain because a rigid turn leaves every inter-satellite distance unchanged.



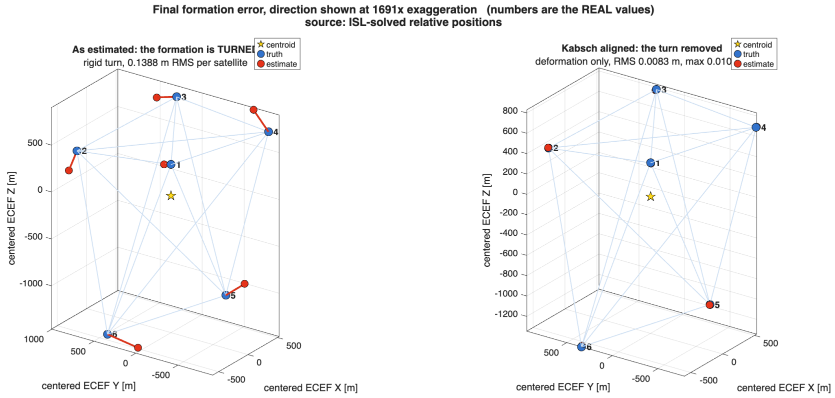
Federated Swarm, Relative Position Error to the Reference Satellite

One curve per satellite: its baseline-vector error against asset 1, the satellite this report is written about. This is the relative counterpart of the per-satellite absolute plot above, and it is the quantity coherent beamforming depends on – a common displacement of the whole formation cancels out of it by construction. Curves are a running median, so the trend is visible without the per-epoch measurement noise; the dashed line is the settled median across all baselines.



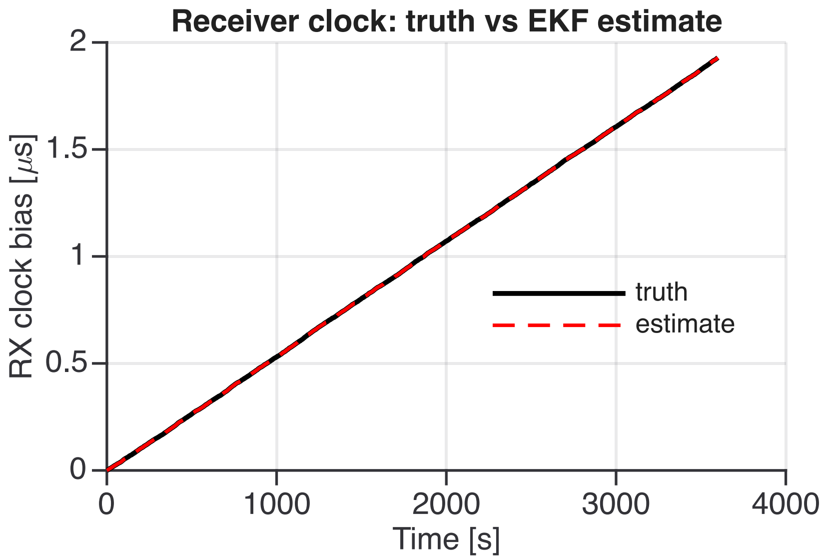
Federated Swarm, Relative Position Error to the Reference, SWARM FRAME

Per-satellite baseline error against asset 1 with the rigid turn removed. Read against the Earth-frame version above: the gap between the two is the formation's orientation error, and it is the term that limits coherent beamforming once the crosslink is calibrated.



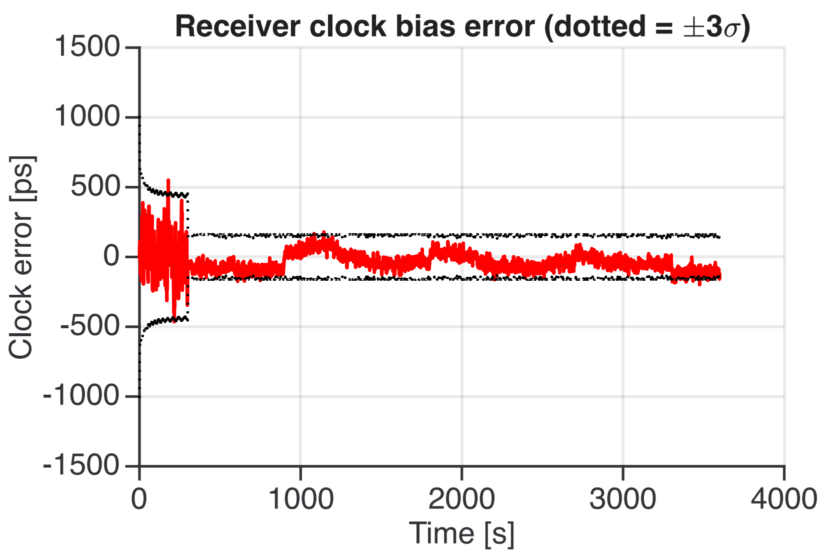
Federated Swarm, Final Formation Error, Direction Exaggerated

Two panels, both centred on the formation centroid and both with the error EXAGGERATED so its DIRECTION is visible; the quoted RMS and max are the real values. Left: the estimate as it stands, which shows the rigid TURN. Right: the same estimate after no-scale Kabsch alignment, so the turn is removed and only DEFORMATION remains. At true scale a sub-degree turn over a kilometre-class formation is invisible, which is why the direction is shown amplified rather than a formation that merely looks correct.



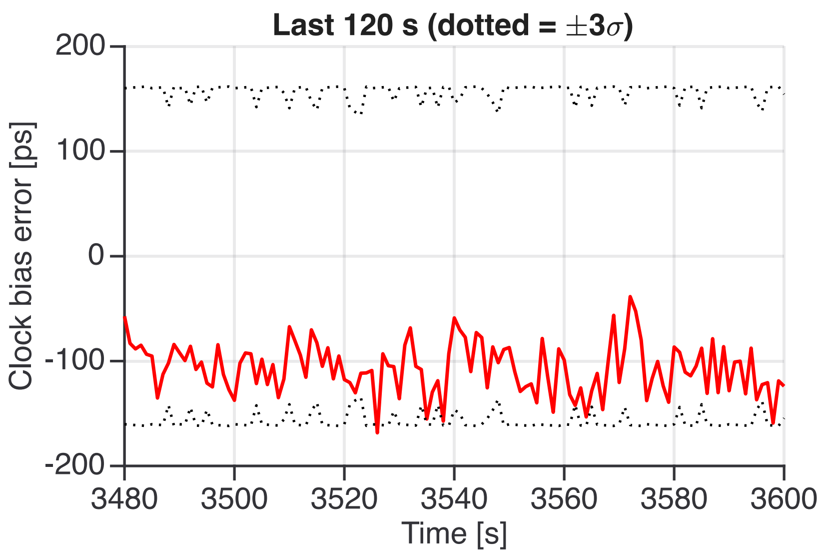
Receiver Clock: True Development vs EKF Estimate

The true receiver-clock bias (black) against the EKF-estimated/corrected clock (red dashed). The difference is the clock-synchronisation error shown next.



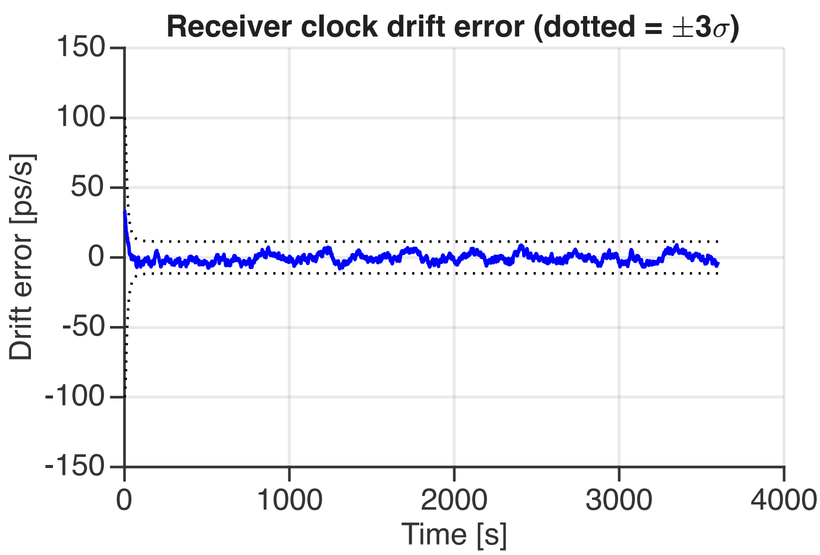
Clock Synchronisation Error

The EKF clock-bias estimation error (solid) against the filter's own $\pm 3\sigma$ envelope (dotted), both in the time domain. The envelope is $3\sqrt{P_{bb}}/c$ taken from the same posterior covariance row the error is formed from, so error and band are always the same state. The clock process follows the selected oscillator power-law noise model (Brown-Hwang two-state). Receiver clock sign is POSITIVE.



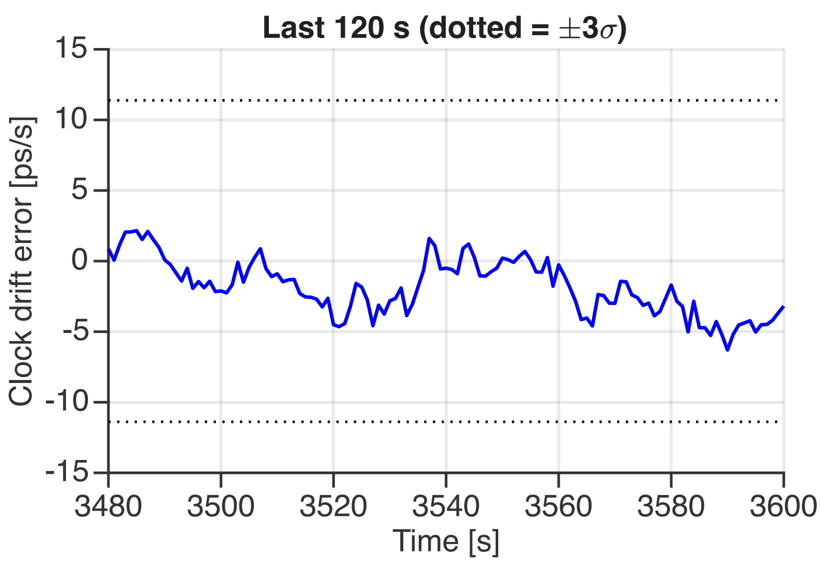
Clock Bias Error, Final 120 s Zoom ($\pm 3\sigma$)

Clock bias estimation error over the final 120 s. RMS over this window: 0.109 ns.



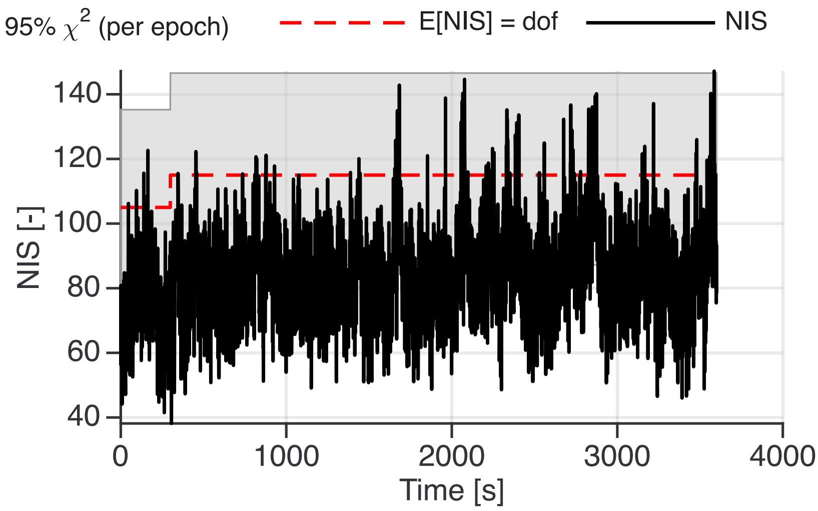
Clock Drift / Fractional Frequency

The receiver clock drift shown as fractional frequency (drift / c) in the time domain (s/s, auto-scaled to ns/s, ps/s ...), against the filter's $\pm 3\sigma$ envelope from the drift covariance row. Brown-Hwang two-state process noise model.



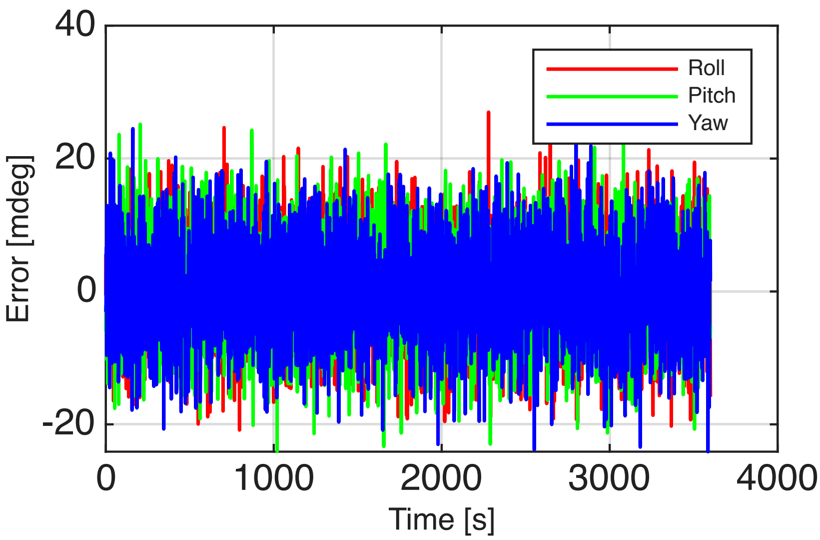
Clock Drift Error, Final 120 s Zoom ($\pm 3\sigma$)

Clock drift estimation error over the final 120 s. RMS over this window: 2.69 ps/s.



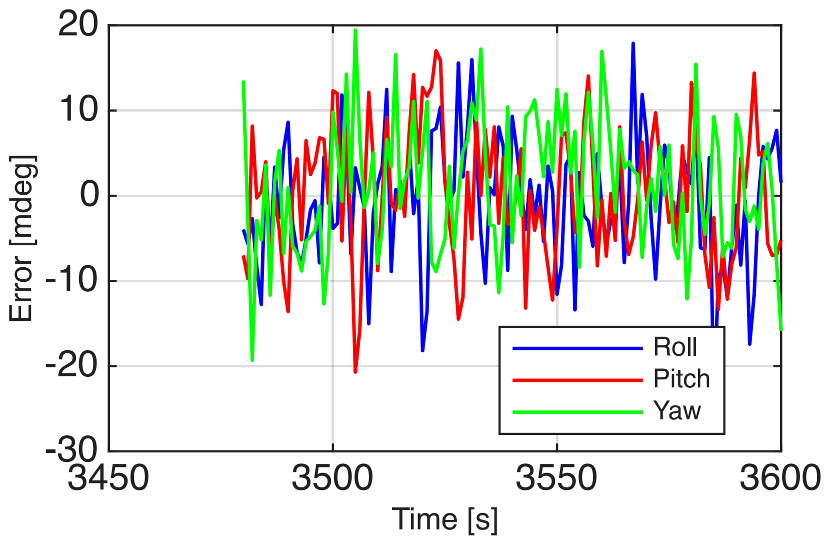
Normalised Innovation Squared

The plotted statistic is the RAW $\nu^\top S^{-1}\nu$, so the value to compare against is NOT one: it is the number of EKF measurement rows in that epoch, here a median of 115 (the red dashed line). Grey is the two-sided 95% chi-square band for a single epoch. Read it by the ratio NIS/dof: about 1 means the filter's R and Q match the errors it is actually seeing; persistently above 1 means the filter is OVERCONFIDENT (its covariance is too small for the residuals it gets); persistently below 1 means UNDERCONFIDENT (covariance inflated, information being thrown away). This run's median NIS/dof is 0.71. NIS is a chi-square consistency test only when the measurement noise is stochastic, zero-mean, and correctly represented in R. In a deterministic validation run it is a numerical conditioning diagnostic and nothing more.



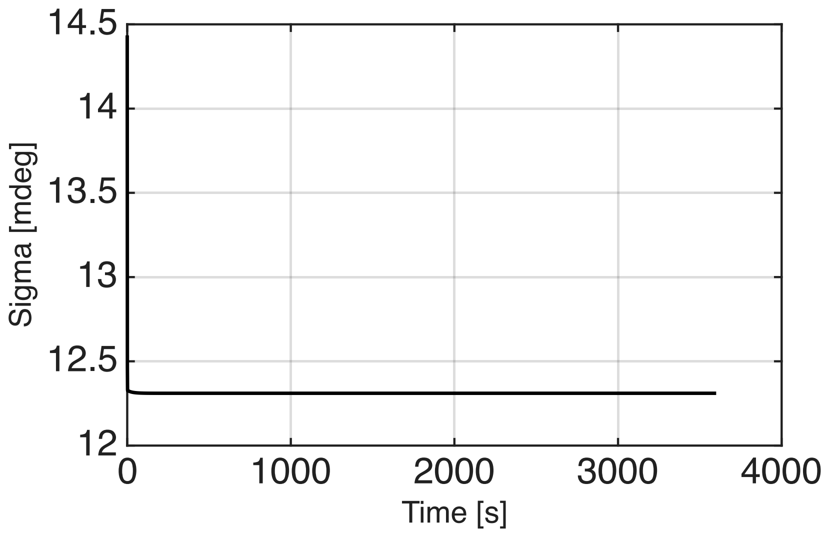
Attitude Error Components: Roll, Pitch, Yaw

Roll, pitch, and yaw estimation errors from diagnostic truth history. If absent, attitude history or plot export was unavailable.



Attitude Components, Final 120 s Zoom ($\pm 3\sigma$)

Roll, pitch and yaw errors over the final 120 s. RMS over this window: roll 0.00763 deg, pitch 0.00761 deg, yaw 0.00728 deg.

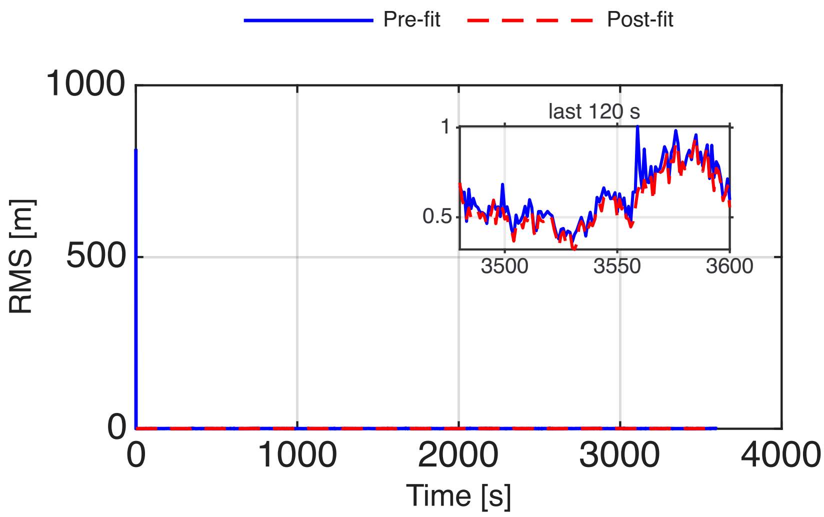
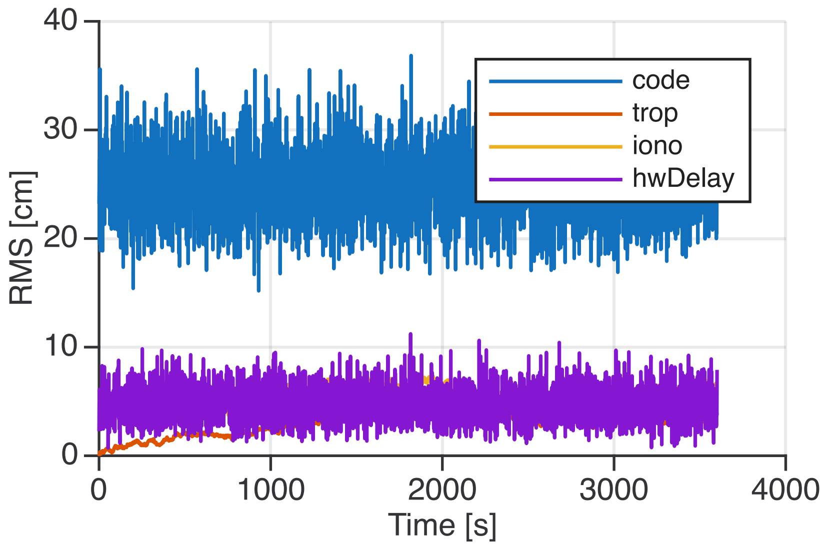


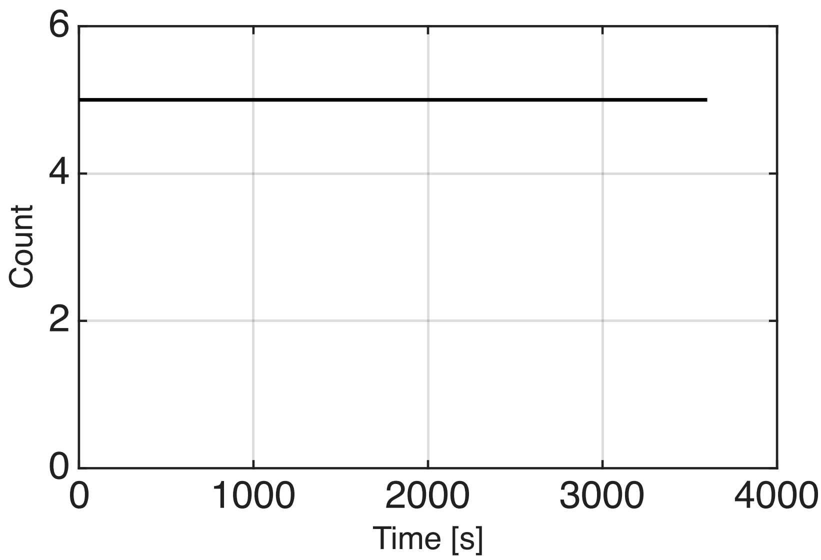
Attitude Covariance Sigma

Square-root sum of the EKF Euler-angle covariance diagonal, in degrees: 0.0144 deg at the first epoch, 0.0123 deg at the last. What to expect: a curve that falls during the opening minutes and then settles flat means the attitude states are observable and the filter has converged. A curve that stays flat from the first epoch means those states received no information and are still sitting on their initial covariance; a curve that grows means process noise is being added faster than the measurements remove it. Read it together with the roll/pitch/yaw errors above: this sigma is what the filter BELIEVES, and it is trustworthy only if those errors stay inside three times it.

3 Measurement and Geometry Validation

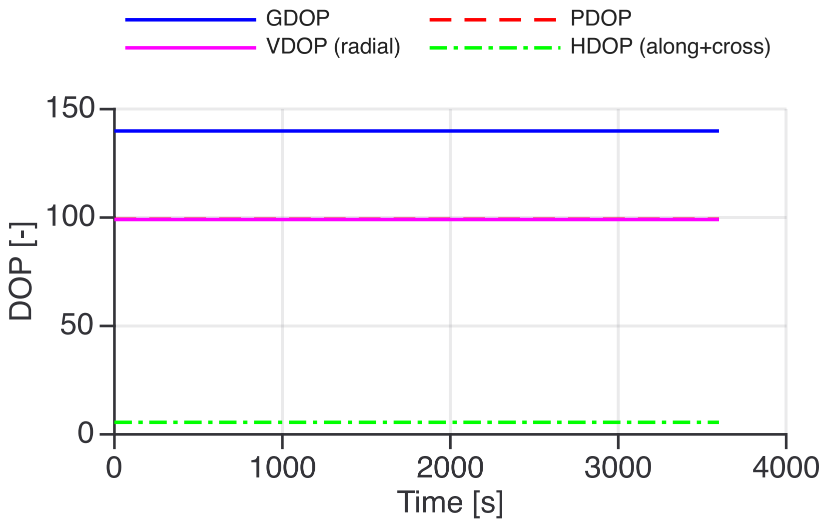
Pre-fit residuals test the predicted measurement model before correction. Post-fit residuals show how much error remains after the EKF update. RMS residuals are useful diagnostics, but they are not by themselves a proof of statistical consistency. In deterministic or partly deterministic validation runs, NIS should be interpreted as a numerical conditioning and model-coupling diagnostic.

Plot	Description and statistical approach
	**EKF Measurement Pre-Fit and Post-Fit Residual RMS** The pre-fit innovation (before the EKF correction) and the post-fit residual (after the update) for the active measurement stack. This is a geometry, clock-state coupling, and convergence diagnostic. The inset repeats the last 120 s on its own scale, because the opening transient otherwise compresses the settled level into the axis. Whole run: pre-fit 13.6 m, post-fit 0.485 m. Last 120 s: pre-fit 0.64 m, post-fit 0.6 m. Post-fit below pre-fit is the update doing work; the two converging on each other means the remaining error is not something the current state vector can absorb.
	**Error Source Contributions** Breakdown of the raw code error into physical contributions: code (geometric and clock residual), troposphere truth-model mismatch, ionosphere truth-model mismatch.



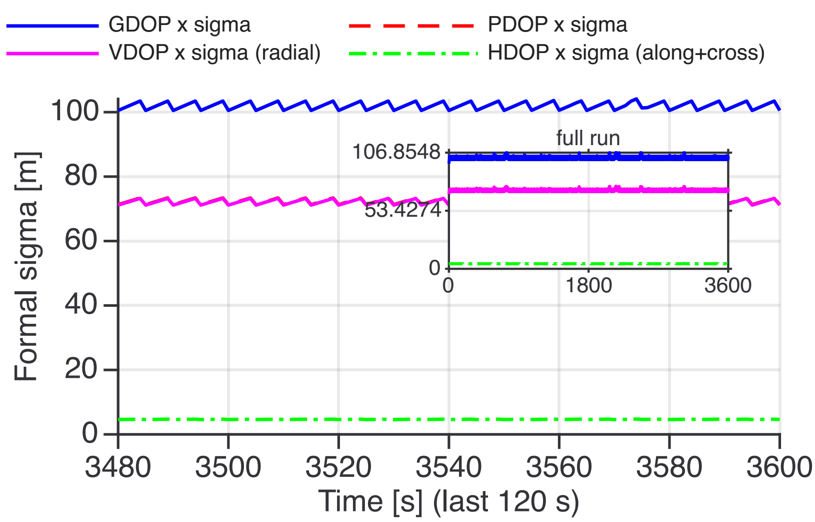
Tower Rows Used by the EKF

The number of ground towers passing the elevation mask at each epoch. Elevation mask applied per-tower before including in the EKF update.



Geometry: unit-weight DOP (dimensionless)

How much the geometry multiplies ranging noise. GDOP covers position and clock, PDOP position alone; VDOP is the radial axis and HDOP along plus cross track. From GEO the ground network sits in a few degrees of sky, so the values run into the hundreds: weak geometry, not unobservable. Flat, because the sight lines barely move. PDOP draws underneath VDOP, which owns nearly all of it.



Weighting: the same DOPs times sigma [m]

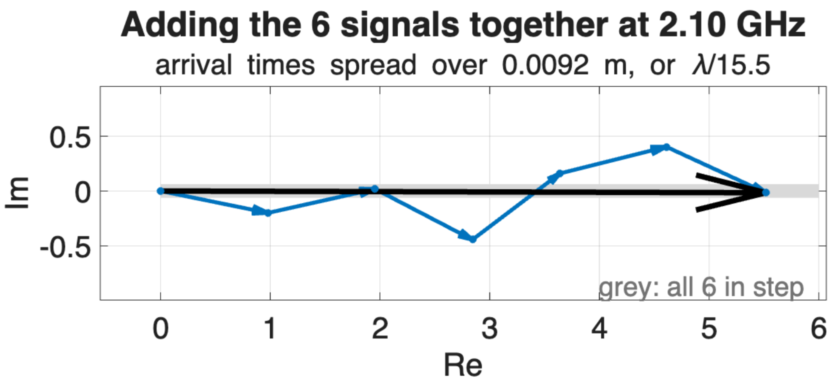
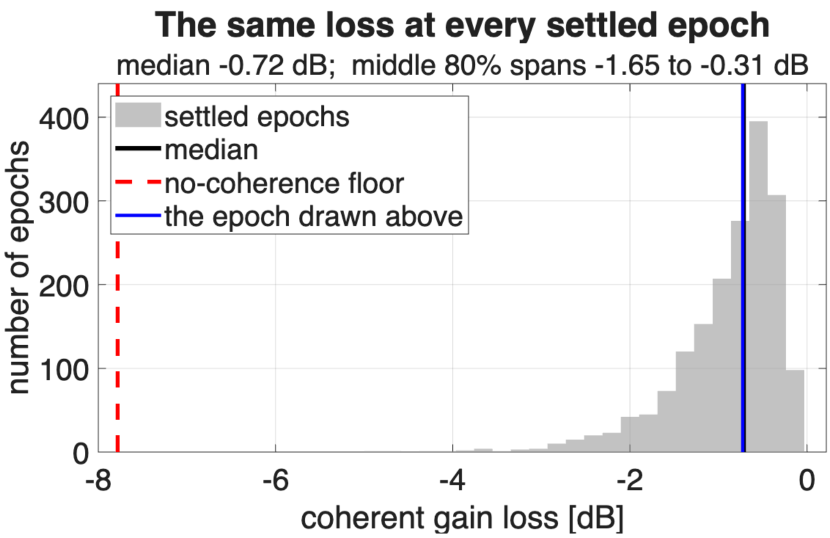
The same four quantities formed with the measurement covariance instead of unit weights, so these are single-epoch formal sigmas in metres. The sawtooth is the tower-clock correction sigma inside R, which grows with the age of the last clock product and resets each update interval. Movement here with the figure above flat is a weighting change, not a geometry change. Inset: the tail at a scale where one cycle is visible.

3.1 Ground-to-Space Geometry and DOP Metrics

Dilution-of-precision factors scale measurement noise into state uncertainty (lower is better). Values are run medians computed from the pseudorange observation geometry in the ECEF line-of-sight frame. This chapter is always shown, independent of any pass/fail gate. A GEO spacecraft viewing a compact cluster of ground transmitters sees nearly parallel lines of sight, so the DOP values are large (hundreds) and nearly constant: the geometry is observable (full rank) but weak.
Both flavours are listed because they are not interchangeable. Unit weight is the classical inv($G^T G$) DOP, dimensionless and a function of the sight lines alone. R-weighted is inv($H^T R^{-1} H$), whose square roots are formal sigmas in metres, so it also moves when only the weighting moves. Quoting the second as a DOP overstates the geometry by roughly the ranging sigma.

Geometry metric	**Value (run median)**
Visible ground transmitters	5.0
Unit-weight DOP (dimensionless, geometry only)	
GDOP (position + clock)	139.92
PDOP (position)	99.24
TDOP (receiver clock)	98.63
VDOP (radial) / HDOP (along+cross)	99.09 / 5.56
R-weighted, i.e. single-epoch formal sigma [m]	
GDOP $\times\sigma$ (position + clock) [m]	102.00
PDOP $\times\sigma$ (position) [m]	72.36
TDOP $\times\sigma$ (receiver clock) [m]	71.89
VDOP (radial) / HDOP (along+cross) [m]	72.21 / 4.66
Axis split (identical for both flavours)	
VDOP / PDOP (radial share of the position dilution)	0.9979 (min 0.9979, max 0.9980)
HDOP / VDOP (along+cross against radial)	0.0645 (min 0.0639, max 0.0653)
Position–clock normal-matrix condition number	1.13e+06
Geometry convention	ECEF line-of-sight; per-tower elevation mask applied

4 Coherent Beamforming Phasor Diagram

Plot	Description and statistical approach
	**Adding the signals up at 2.10 GHz** One arrow per spacecraft, all the same length, each turned by how early or late its signal arrives – a full turn per wavelength. Laid end to end, the black arrow is what reaches the ground. In step they would stretch out to N; spread round the circle they cancel, and no re-pointing recovers them. Drawn epoch: path spread 9.2 mm. A delay shared by all N is removed first, because it steers the beam without costing gain.
	**How much that total moves between epochs** The chain above is a single epoch, and a single epoch is a draw rather than a measurement: once the arrival times spread past a fraction of a wavelength the total wanders by several dB. Here is the same loss at every settled epoch – black the median, blue the epoch drawn above, red the floor $20\log_{10}(1/\sqrt{N})$ that N unsynchronised spacecraft would give. Measured: median -0.72 dB, middle 80% -1.65 to -0.31 dB, floor -7.78 dB. Quote the median.

5 Scientific Verdict

This section summarises estimator performance against the known synthetic truth. The final-sample values are endpoint diagnostics; do not use them alone to rank performance. Run-level statistics below cover the full record and the final window. All results are for the controlled synthetic scenario only and are not a real-data or PPP-grade performance claim.

Quantity	**Value**
Final 3D position estimation error	0.077622 m
Final receiver clock estimation error	-0.037161 m
Final clock range-equivalent error (ps)	-124.0 ps
Final pre-fit measurement innovation RMS	0.597393 m
Final post-fit measurement residual RMS	0.555208 m
Max EKF measurement rows / epoch	115
Configured code rows / epoch	40
Configured Doppler rows / epoch	20
Configured carrier rows / epoch	40
Carrier slip detector	model-step-compensated residual jump
Product epoch boundary events	0 compensated / 0 total
Confirmed carrier slips	No confirmed slips in nominal run
False product-boundary resets	0
Covariance mode	block covariance for shared tower-clock product
Code block covariance	fallback: diagonal only
Carrier R policy	Shared product bias and drift blocks
Doppler R policy	Tracking only plus block
R symmetric positive-definite	yes
Mean NIS (raw, all rows)	82.83
NIS dof (EKF rows / epoch)	114.2 (mean)
Mean NIS / dof (expected 1.00)	0.725
95% χ^2 band, dof $= 114.2(mean)$	[86.5, 145.6] raw
same band per dof	[0.758, 1.276]

The χ^2 band is the two-sided 95% interval for a **single** epoch. The means above are averaged over epochs whose NIS is autocorrelated in this simulation, so the band bounds the per-epoch scatter and is not a hypothesis test on the mean.

Metric	**Full RMS**	**Final sample**
3D position error [m]	1.197621 m	0.077622 m
Clock error [m]	0.025295 m	-0.037161 m
Pre-fit innovation RMS [m]	n/a	0.597393 m
Post-fit residual RMS [m]	n/a	0.555208 m

6 Federated Swarm (N independent single-asset EKFs + ISL/TWSTFT relative layer)

This scenario is a *federated* swarm of 6 satellites. Each satellite estimates its own absolute state from the ground towers with its *own* single-asset EKF (no chief, no shared covariance, design decision D1), so no filter can drag another into divergence. **The detailed report above is the reference satellite's (asset 1, the "chief") *own single-asset run***; this swarm comprises 6 such satellites, reported symmetrically in the tables below. In this configuration, the synthetic two-way ISL / sat–sat TWSTFT relative layer is a read-only diagnostic post-processor and is never fused back into any per-asset absolute solution.

Per-satellite absolute estimate (each from its own EKF)

asset	absErr [m]	absSig [m]	err/σ	clkErr [ns]	relPos raw [m]	relPos solved [m]
1*	0.096	0.009	11.05	1749.304	0.000	0.0000
2	0.112	0.009	12.89	1749.275	0.055	0.0701
3	0.052	0.009	5.99	1749.236	0.087	0.0369
4	0.044	0.009	5.06	1749.182	0.071	0.0701
5	0.073	0.009	8.41	1749.224	0.065	0.0919
6	0.094	0.009	10.84	1749.186	0.169	0.1241

*reference asset. **relPos raw** = each satellite's relative-position error vs the reference from differencing the two *independent* EKF absolutes (pre-ISL). **relPos solved** = the same quantity *after* the ISL free-network shape solve. Both retain the native estimate-frame gauge (ref-differenced, with no truth-based frame alignment). The solved result nevertheless consumes truth-synthesized synthetic observations. Tail-averaged (last 20 %).

Relative layer (ISL shape / sat–sat TWSTFT clocks)

Weakly observable: no (the ISL shape-solve normal matrix is well-conditioned).

quantity	error	formal σ	
formation baseline error (raw, pre-ISL)	0.031 m	–	–
formation baseline error (solved)	0.0053 m (0.53 cm)	–	–
best-fit-rigid shape error (solved)	0.0111 m (1.11 cm)	0.0204 m	
relative clock (sat–sat TWSTFT)	off	–	–
NIS	not applicable: the shape/clock layer is a per-epoch least-squares solve, n		

The per-satellite absolute-error, relative-layer, and Kabsch alignment plots appear with the state-estimation figures above, immediately after the RAC final-zoom plot.
Kabsch alignment uses truth as the reference frame and is a shape-only diagnostic; it is not fed back into any estimate.

Each satellite's absolute is wall-limited (radial–clock common mode, err/σ near 1–3), because reverse-GNSS towers only weakly separate radial position from the receiver clock. The ISL/TWSTFT relative layer cannot re-open that wall (no shared covariance), yet it recovers the observable formation shape and relative clocks.